Aakash Karthikairaja

AI / ML Engineer | LLMs, RAG, Evaluation

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SUMMARY

AI/ML engineer with 2 years across LLM data operations, RAG pipelines, and analytics. Hands-on with LLM training-data curation (Claude, Amazon Nova), embedding generation, hallucination evaluation, and production-style RAG systems built on LangChain and vector databases. M.Tech in Artificial Intelligence (SRM IST, 2026); conference-published applied ML research.

PROFESSIONAL EXPERIENCE

Machine Learning Data Associate - II **03/2025 – 11/2025**
Amazon Development Centre Chennai

- Contributed to LLM training-data pipelines supporting Amazon Nova and Claude model datasets, curating high-quality conversational data under a structured taxonomy at multi-million-token scale.
- Built and optimized RAG workflow components (LangChain, vector databases, embedding generation), contributing to a 40% reduction in document retrieval latency and improved context relevance.
- Implemented programmatic integrity checks and automated validation for model outputs, contributing to a 22% reduction in hallucinations and removal of critical edge-case errors.
- Developed Python data-processing and benchmarking workflows that accelerated model performance evaluation by 35%.
- Automated data ingestion with GitLab CI and Docker, shortening retraining deployment cycles by over 2 weeks.
- Partnered with Core ML Engineers to define evaluation benchmarks for LLM quality and hallucination measurement.

Associate Developer **10/2023 – 01/2025**
Brainvoice.ai Chennai

- Architected and deployed AI-based conversational intelligence and speech analytics workflows using Python and scalable cloud infrastructure, significantly optimizing real-time natural language processing capabilities.
- Engineered structured backend analytical pipelines and integrated complex data models, partnering with cross-functional engineering teams to optimize supply chain execution and operational processes.
- Executed high-impact, data-backed targeting strategies efficiently and deployed ML-driven algorithmic workflows that successfully decreased lead generation expenses by 96%.
- Streamlined cloud operational processes and modernized legacy reporting systems alongside senior engineers, reducing overall operational overhead and compute costs by 30%.

EDUCATION

Master of Technology | Artificial Intelligence **08/2023 – 05/2026**
SRM Institute of Science and Technology Chennai

Bachelor of Engineering | Mechanical Engineering **09/2019 – 06/2023**
Sri Krishna College of Technology Coimbatore

SKILLS

Generative AI & Large Language Models (LLMs)
- Retrieval-Augmented Generation (**RAG**)
- LangChain, LlamaIndex
- Model Context Protocol (**MCP**), agentic workflows
- Prompt and context engineering
- LoRA/PEFT fine-tuning
- Models: Claude 3.5/3.7, Amazon Nova, Llama 3

Backend & Data Stores
- Python (advanced), SQL, Bash
- FastAPI, Flask, RESTful APIs, microservices
- PostgreSQL, MongoDB, SQLite

Cloud Computing, MLOps & Infrastructure
- AWS (S3, EC2, Lambda)
- GCP; Docker
- GitLab CI, GitHub Actions
- MLflow, model versioning, registry and monitoring
- IAM, RBAC

Data Engineering
- ETL pipeline architecture
- Pandas, NumPy
- Foundational exposure to Hive, MapReduce, Sqoop

Machine Learning & Deep Learning
- PyTorch, TensorFlow, Scikit-Learn, XGBoost
- CNN, RNN, LSTM, Transformer architectures
- Time-series forecasting, decision-level fusion, SHAP interpretability
- OpenCV, NLP text processing

Evaluation & Guardrails
- Ragas, DeepEval, hallucination benchmarking
- Output validation frameworks, programmatic integrity checks
- Privacy guardrails

Retrieval & Vector Search
- Pinecone, Milvus, pgvector
- Embedding generation, semantic retrieval
- Retrieval-latency optimization

PROJECTS

Healthcare Agentic AI Multiturn Conversational
First500Days
- Built a multi-turn RAG chatbot delivering context-aware maternal and infant care guidance grounded in indexed clinical guidelines.
- Implemented semantic retrieval over a vector database for low-latency, high-fidelity responses.
- Added data-privacy guardrails and output-validation pipelines to limit hallucinations in a healthcare setting.

AI-SaaS Platform Development
Dream Reflection Media
- Built a cloud-based AI SaaS platform with custom ML models served through low-latency inference APIs.
- Optimized backend data pipelines for concurrent request handling; set up containerized CI/CD deployment.

Freight Load-Matching Platform (Founder / Builder)
Python, REST APIs, PostgreSQL, geolocation
- Designing and building a broker-free logistics marketplace for India that connects shippers with truck operators, targeting reduced empty-return trips and lower brokerage friction.
- Implementing load listings (cargo type, capacity, route), driver-side nearby-load discovery, bidding flow, and shipper-side truck search.
- Integrating location data and basic GPS-based load tracking for shippers.

PUBLICATIONS

Hybrid LSTM-XGBoost Framework Integrating Weather Forecast Inputs for Water-Optimal Tank and Pump Decisions **04/2026**
Published Researcher | NIT Tiruchirappalli
- Authored and presented applied machine learning research at the International Conference on Advances in Green, Net Zero Innovation (**AGNI-S 2026**).
- Designed a decision-level fusion model (LSTM + XGBoost) with an 18% reduction in prediction error versus standalone baselines; applied lag features, rolling averages, and SHAP interpretability.
- Deployed an end-to-end forecasting web application (Flask, SQLite) with real-time data ingestion for irrigation scheduling.